

Quick Recall: Key Definitions

Ordering (Ch. 2.1). A relation $\preceq$ on a set X is a (partial) *ordering* if it is reflexive, antisymmetric, and transitive. A (partial) ordering is a *linear ordering* if for every pair $x, y \in X$, either $x \preceq y$ or $y \preceq x$ holds. As a relation, $\preceq$ is a subset of $X \times X$.

Linear extension (Thm. 2.2.1). Let $(X, \preceq)$ be a finite partially ordered set. Then there exists a linear ordering $\leq$ on X such that $x \preceq y$ implies $x \leq y$. (Such $\leq$ is called a *linear extension* of $\preceq$.)

Complete bipartite graph (§4.7). For disjoint sets X and Y , $K_{X,Y}$ denotes the graph with vertex set $X \cup Y$

and edge set $\{\{x, y\} : x \in X, y \in Y\}$. Setting $a = |X|$, $b = |Y|$, this graph is isomorphic to $K_{a,b}$. It contains no triangle and has ab edges.

Turán's theorem (Thm. 4.7.1). For every n , $T(n) = \lfloor n^2/4 \rfloor$, where $T(n)$ is the maximum number of edges of a triangle-free graph on n vertices. Equality is attained uniquely (up to iso.) by $K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}$ (Thm. 4.7.2).

Edge-connectivity (§4.6). A graph G is *k-edge-connected* if we obtain a connected graph by deleting any $k - 1$ edges of G .

Practice: Textbook Theorem 4.7.3 — Degree-dominating bipartition for triangle-free graphs

Let $G = (V, E)$ be a triangle-free graph. Prove that there exists a partition of V into two subsets X and Y such that for every vertex $x \in V$,

$$\deg_G(x) \leq \deg_{K_{X,Y}}(x).$$

Remarks

One clever choice of X, Y does the whole job — no induction needed. Try:

- Pick $x_0 \in V$ of *maximum* degree in G . Let $Y = N_G(x_0)$ and $X = V \setminus Y$.
- For $x \in X$: what is $\deg_{K_{X,Y}}(x)$, and why is it $\geq \deg_G(x)$?
- For $y \in Y$: use that G is triangle-free to argue that Y is an *independent* set in G . What does this say about $N_G(y)$?

Corollary (Turán, Thm. 4.7.1). Summing the inequality gives $|E(G)| \leq |E(K_{X,Y})| \leq \lfloor n^2/4 \rfloor$.

Practice: Textbook Exercise 2.4.1(b)* — Partial orderings as intersections of linear orderings

Prove that every partial ordering $\preceq$ on a finite set X can be expressed as the intersection of linear orderings of X .

Remarks

Useful lemma (part (a) of Ex. 2.4.1). If $\leq_1, \dots, \leq_k$ are orderings on X , then $\bigcap_{i=1}^k \leq_i$ is again an ordering on X (check reflexivity, antisymmetry, transitivity from the definitions).

So the intersection of any family of linear extensions of $\preceq$ is automatically an ordering containing $\preceq$. The work is to choose *enough* linear extensions so that no extra pairs sneak into the intersection: for every incomparable pair $\{a, b\}$ in $\preceq$, build one linear extension with $a < b$ and another with $b < a$.

Practice: Textbook Exercise 4.6.6* — Toughness of d -regular, d -edge-connected graphs

Let $d \geq 3$ and let G be a d -regular (every vertex has degree d), d -edge-connected graph. Prove G is *tough*: for every $k \geq 1$ and every set $S \subseteq V(G)$ with $|S| = k$, the graph $G - S$ has at most k connected components.

Remarks

Count edges between S and the components of $G - S$. Each component contributes at least d edges to the cut (by d -edge-connectivity), but S only has dk endpoints available in total. Handle the small case carefully.